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Abstract

This report presents the ongoing development of a network vulnerability assessment tool designed to enhance cybersecurity measures in the face of rapid advancements in networking technology. The tool aims to identify vulnerabilities, providing a crucial line of defense against cyber attackers. It leverages the latest advancements in cybersecurity to prevent data breaches, system disruptions, and network failures. The tool also addresses the global security concern posed by weak passwords, providing recommendations for improving password security. The report provides an overview of the project's progress, including completed tasks and future plans.

Table of Contents

1	Introduction	1
1.1	Problem Statement	2
1.2	Project as a Solution	3
1.3	Aim and Objectives	3
2	Background	5
2.1	Client's Name and Description	5
2.2	Client Requirements.....	5
2.3	Understanding the Project.....	5
2.3.1	Project Elaboration	5
2.3.2	Project Deliverables	6
2.4	Similar Tools	7
2.4.1	Nessus.....	7
2.4.2	Nmap.....	8
2.4.3	Sparta.....	8
2.5	Comparison Table	9
3	Development.....	10
3.1	Considered Methodology	10
3.1.1	Waterfall Methodology	10
3.1.2	Agile Methodology	11
3.2	Selected Methodology.....	12
3.2.1	Evolutionary Prototyping.....	12
3.3	Work Breakdown Structure	15
3.4	Gantt Chart.....	15
4	Analysis.....	16
4.1	Progress Table.....	16
4.2	Progress Review	16
5	Future work	18
6	References.....	20
7	Bibliography	22
8	Appendix.....	23
8.1	Survey Findings	23
8.2	Milestone.....	30

8.3	Project's GUI Design	31
8.4	Development	32
8.4.1	Login GUI	32
8.4.2	Register GUI	33
8.4.3	Dashboard GUI	33
8.5	Development Testing	34
8.5.1	Registering Username and Password	34
8.5.2	Login	34
8.5.3	Host Discovery Button	35
8.5.4	Port Scanning Button	35
8.5.5	Generate Report Button	36
8.5.6	Report Button	36
8.6	Client approval Letter	38

Table of Figures

Figure 1: Nessus GUI (Tenable®, 2023).....	7
Figure 2: Sparta GUI	9
Figure 3: Waterfall Methodology (Ardhiansyah, et al., 2022).....	11
Figure 4: Agile Methodology (Grepon, et al., 2021).....	12
Figure 5: Evolutionary Prototyping (Lapada, 2019)	14
Figure 6: Work Breakdown Structure (WBS) of the project	15
Figure 7: Gantt Chart of the project.....	15
Figure 8: Progress table of the project	16
Figure 9: Survey Question 1.....	24
Figure 10: Survey Question 2.....	24
Figure 11: Survey Question 3.....	25
Figure 12: Survey Question 4.....	25
Figure 13: Survey Question 5.....	26
Figure 14: Survey Question 6.....	26
Figure 15: Survey Question 7.....	27
Figure 16: Survey Question 8.....	27
Figure 17: Survey Question 9.....	28
Figure 18: Survey Question 10.....	28
Figure 19: Survey Question 11.....	29
Figure 20: Survey Question 12.....	29
Figure 21: Milestone.....	30
Figure 22: Login Design	31
Figure 23: Register Design.....	31
Figure 24: Dashboard Design	32
Figure 25: Login Interface	32
Figure 26: Register Interface.....	33
Figure 27: Dashboard Interface.....	33
Figure 28: Registering Username and Password	34
Figure 29: Login Using Login Credentials	34
Figure 30: Host Discovery on the Network.....	35

Figure 31: Port Scanning on the Network.....	35
Figure 32: Generating Report.....	36
Figure 33: Report Lists	37
Figure 34: Pdf File	37
Figure 35: Client Approval Letter	38

Table of Tables

Table 1: Comparison table 9

1 Introduction

The rapid advancement of the internet and communication technologies has led to a significant increase in the number of network access frameworks worldwide. These frameworks serve as the backbone for our increasingly interconnected world, enabling seamless communication and data exchange across geographical boundaries (Pandit, 2021). As technology continues to evolve, these network frameworks are being leveraged to develop a plethora of new applications. These applications, which range from social media platforms to financial systems, rely heavily on these networks and the internet to transmit vast amounts of user data. The volume of this data has seen an exponential increase, mirroring the rapid pace of technological innovation (Pandit, 2021).

However, this progress has not been without its challenges. One of the most pressing issues is that vulnerabilities have emerged as a major source of cyber threats in the modern era of computing. The assessment of vulnerabilities in networking devices is crucial for maintaining a secure network. Unattended vulnerabilities in the network present immense possibilities for exploitation (Muralidharan, et al., 2019). Default or weak usernames and passwords for devices can be exploited by attackers, further emphasizing the need for robust security measures. Software companies and government organizations are diligently working on various vulnerability assessment tools and methodologies to reduce security risks in networks.

In response to this growing concern, the network vulnerability assessment (NVA) tool comes into play. The tool follows the process of identifying security vulnerability in the systems, quantifying and analysing them (Strahinja, 2023). This process allows for a comprehensive understanding of the network's security posture. The NVA tool not only identifies vulnerabilities but also ranks them based on their severity, providing a clear roadmap for remediation. By doing so, it enables organizations to prioritize their efforts and resources effectively, focusing first on the most critical vulnerabilities that pose the greatest risk. Furthermore, the tool provides detailed reports and actionable insights, aiding in the development of robust security strategies.

1.1 Problem Statement

The rapid advancements in networking technology have brought about a multitude of benefits, from increased connectivity and communication speed to enhanced data sharing capabilities. However, these advancements also come with new and undetected vulnerabilities. Cyber attackers are constantly on the lookout for these vulnerabilities, aiming to exploit them for malicious purposes. When an attacker successfully exploits a vulnerability, it can lead to significant damage. This includes data breaches, system disruptions, and network failures. In severe cases, these attacks can cause complete system failure, leading to substantial downtime and potential financial loss.

In 2020, it was reported that 87% of organizations experienced an attempted exploit of a known vulnerability (Check Point, 2023). This statistic underscores the prevalence of these types of attacks and the importance of robust network security. One of the most significant attacks that year was the Sunburst attack, which breached thousands of governments and private sector organizations. The Sunburst malware was distributed via compromised updates to the SolarWinds Orion network management software, demonstrating the sophisticated methods that attackers can use to exploit network vulnerabilities.

In addition to these threats, weak passwords pose a significant global security concern, as they can be easily exploited by attackers to gain unauthorized access to networks. When users create passwords that are easy to guess or have been exposed in previous data breaches, attackers can use methods like brute force attacks, credential stuffing, and password spraying to gain access. In fact, 93% of the passwords used in brute force attacks include 8 or more characters.

One of the most common types of attacks that exploit weak passwords is ransomware attacks. In these attacks, cybercriminals encrypt the victim's data and demand a ransom for its release. Researchers have found that brute force attacks have become the most common means of spreading ransomware, overtaking phishing emails and spam.

1.2 Project as a Solution

The project aims to develop a network vulnerability assessment tool to address the pressing issues highlighted above. The tool will be designed to identify and assess vulnerabilities in networking technology, providing a crucial line of defense against cyber attackers. The tool will leverage the latest advancements in cybersecurity to detect vulnerabilities. By doing so, it will help to prevent cyber attackers from exploiting these vulnerabilities for malicious purposes. The tool will be capable of identifying a wide range of vulnerabilities, from those that could lead to data breaches and system disruptions, to those that could result in network failures or even complete system failure.

In addition, the tool will also address the global security concern posed by weak passwords. It will include features to assess the strength of passwords. This will help to prevent attackers from gaining unauthorized access to networks through methods like brute force attacks, credential stuffing, and password spraying.

1.3 Aim and Objectives

The project's aim is to develop a comprehensive network vulnerability assessment (NVA) tool that empowers businesses to proactively detect and mitigate vulnerabilities within a network's architecture and its connected devices. It also enhances the overall security posture of networks, reducing the risk of cyber threats and fortifying the integrity, confidentiality, and availability of digital infrastructure.

The Objectives of the project are:

- Develop a powerful and user-friendly network vulnerability assessment tool.
- Provide cost-effective solutions to the organization.
- Develop a standardized process for conducting in-depth vulnerability assessments.
- Develop a reporting mechanism that generates clear, concise, and actionable reports.
- Integrate with other tools for accuracy of the assessments.

- Develop a system that can check the weak or default username and password on the networking devices.

2 Background

2.1 Client's Name and Description

Name of the client: Mr. Prabin Lal Shrestha

Client's Description: The client for this project is Mr. Prabin Lal Shrestha. He holds the position of CEO and IT Manager at Feather Solutions Private Limited, a company located in Baluwatar, Kathmandu, Nepal. Mr. Shrestha has agreed to be the client for this project, as he believes it aligns well with his professional interests and could be beneficial for his work and company. He is open to collaborating on ideas and is willing to adhere to the established conditions.

2.2 Client Requirements

After collaborating and having several meetings with the client, the client's requirements for this project are as follows:

- User-friendly GUI.
- Generate report of the results.
- Port scanning feature that can able to discovery service and service version of the open ports.
- The tool should be able to find vulnerability by matching service and service version of the open ports to the vulnerability database like CVE database, NVD, etc.
- Discover active host on the network.
- Check for default or weak username and password on the network.

2.3 Understanding the Project

2.3.1 Project Elaboration

The project involves the creation of a 'Network Vulnerability Assessment Tool' using python programming. This tool aids organizations in identifying vulnerabilities in their network devices, thereby enhancing the security of their computer networks. The

tool can discover all the active devices connected to a network using the Python Nmap library. Each device on a network has 'ports' that allow information to flow in and out. The tool checks these ports to determine which ones are open, as open ports can potentially allow unwanted visitors access, and also checks for service and service running on the open port using the Python Nmap library. These services and their versions are then cross-referenced with a vulnerability database (such as the CVE database or NVD) to identify any existing vulnerabilities. The tool also checks for weak or default usernames and passwords used on the network to ensure they are robust enough. Furthermore, the tool generates detailed reports of its findings, providing a comprehensive overview of the network's security status.

2.3.2 Project Deliverables

The primary target audience of this project is small-sized organizations that may not have the financial resources to implement robust security measures to prevent data breaches or attacks. The network vulnerability assessment tool we are developing is designed to be cost-effective, user-friendly, and efficient. It aims to identify potential vulnerabilities in the network infrastructure, providing these organizations with a comprehensive understanding of their security posture. This tool will conduct an in-depth analysis of the network, examining various devices for any potential weaknesses. The results from these assessments will then be compiled into a detailed report, highlighting the identified vulnerabilities.

By offering a solution that is both affordable and effective, I hope to empower small organizations to enhance their network security, thereby reducing the risk of data breaches and attacks. This tool is not just a protective measure, but an investment in the organization's future, ensuring the safety and integrity of its data.

2.4 Similar Tools

2.4.1 Nessus

Tenable Nessus, recognized as the industry's most widely deployed solution for vulnerability assessment, plays a crucial role in helping organizations reduce their attack surface and ensure compliance. Tenable Nessus features high-speed asset discovery, configuration auditing, target profiling, malware detection, sensitive data discovery, and more (Tenable Nessus, 2023).

Tenable Nessus supports more technologies than competitive solutions, scanning operating systems, network devices, hypervisors, databases, web servers, and critical infrastructure for vulnerabilities, threats, and compliance violations. With the world's largest continuously updated library of vulnerability and configuration checks, and the support of Tenable, Inc.'s expert vulnerability research team, Tenable Nessus sets the standard for vulnerability scanning speed and accuracy (Tenable Nessus, 2023).

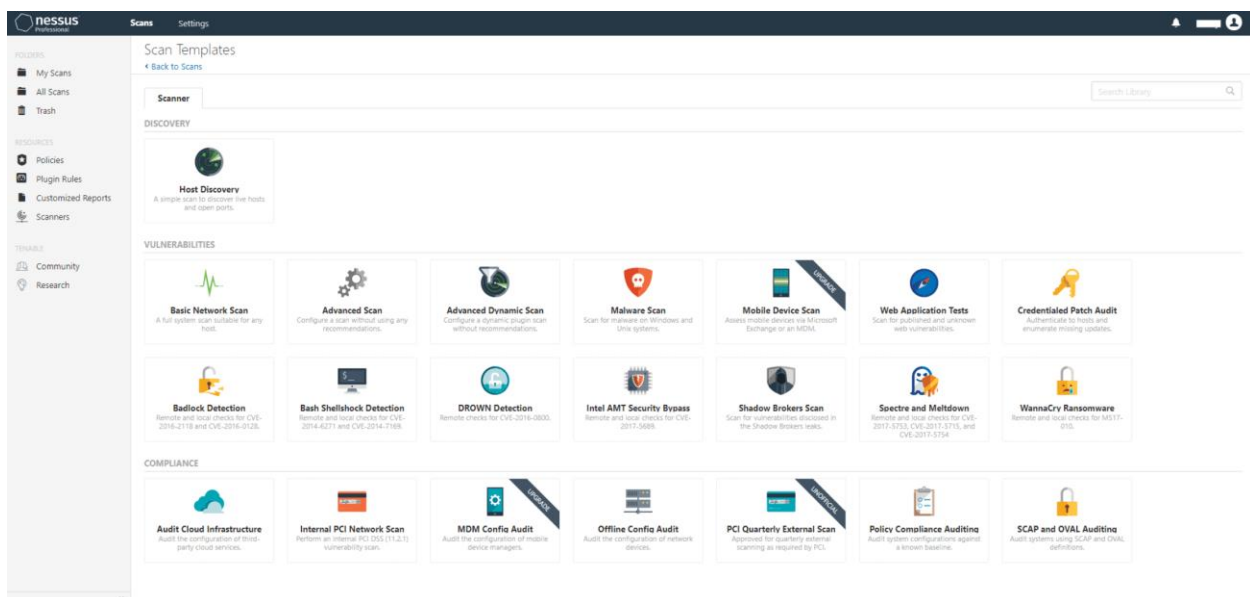


Figure 1: Nessus GUI (Tenable®, 2023)

2.4.2 Nmap

Nmap, also known as Network Mapper, is a free and open-source tool that is widely used for network discovery and security auditing. It is highly valued by system and network administrators for a variety of tasks. These include network inventory management, scheduling service upgrades, and monitoring the uptime of hosts or services (Nmap, 2023).

Nmap employs raw IP packets in innovative ways to ascertain a host of information about the network. It can identify available hosts on the network, the services they offer (including the application name and version), the operating systems they run (along with the OS versions), the type of packet filters or firewalls in use, and many other characteristics. It is designed with the capability to swiftly scan large networks, Nmap is equally effective when used against single hosts. It is compatible with all major computer operating systems, and official binary packages are readily available for Linux, Windows, and Mac OS X (Nmap, 2023).

2.4.3 Sparta

SPARTA is a GUI application, developed with Python, that serves as a network penetration testing tool for Kali Linux. It streamlines the scanning and enumeration phase, delivering faster results (Ganesh, 2023). One of the standout features of the SPARTA GUI toolkit is its ability to scan and detect the service running on the target port. Additionally, it offers a Brute force attack for scanned open ports and services, which forms an integral part of the enumeration phase (Ganesh, 2023).

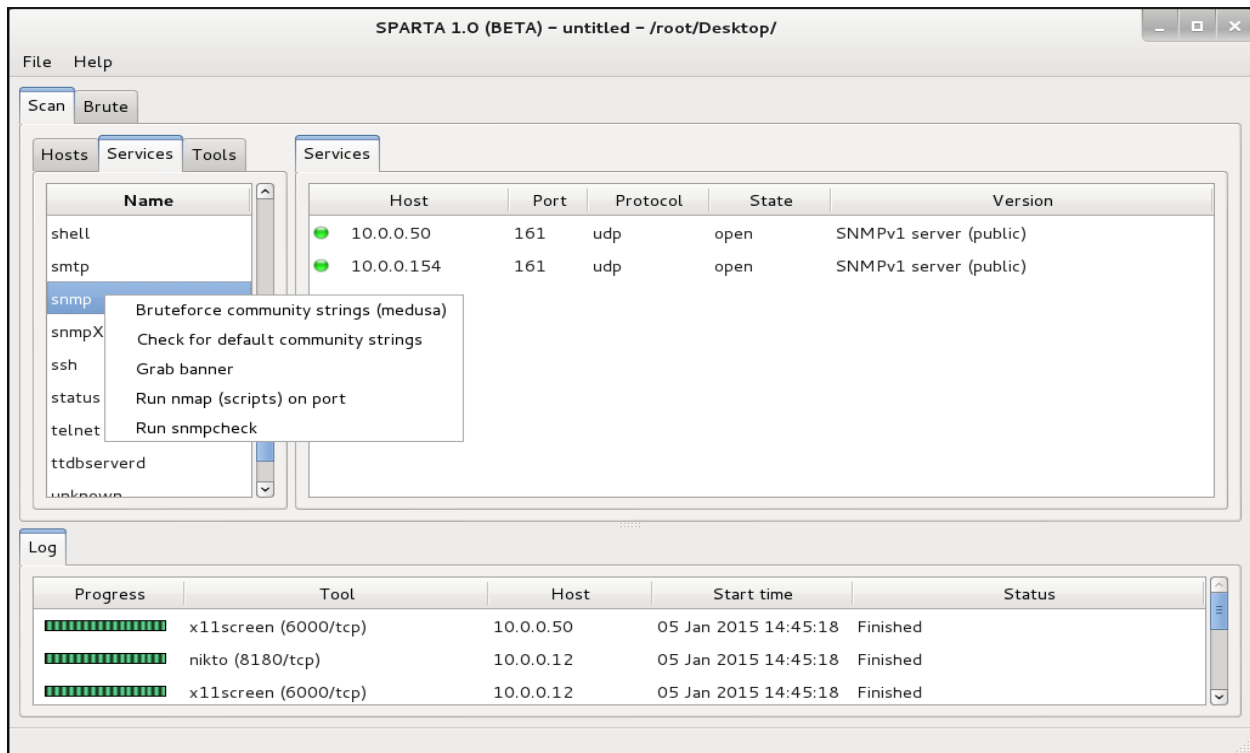


Figure 2: Sparta GUI

2.5 Comparison Table

S.N.	Features	Nessus	Nmap	Sparta	My project-NVA Tool
1	User-friendly	✓	✗	✓	✓
2	Host discovery	✓	✓	✗	✓
3	Port scanning	✓	✓	✓	✓
4	Discover vulnerability	✓	✓	✓	✓
5	Credential Testing	✓	✗	✓	✓
6	Cost-effective	✗	✓	✓	✓
7	Generate Report	✓	✗	✗	✓

Table 1: Comparison table

3 Development

3.1 Considered Methodology

3.1.1 Waterfall Methodology

The Waterfall Model was the first process model to be introduced, often referred to as a linear sequential life cycle model due to the software development process following a linear sequential flow (Senarath, 2021). In this approach, the entire process of software development is divided into distinct phases. Typically, the outcome of one phase serves as the input for the subsequent phase in a sequential manner. The Waterfall Model is particularly well-suited for projects where the objectives are clearly defined from the outset (Senarath, 2021).

The reasons for not selecting the Waterfall methodology for this project are:

- The model follows a linear sequence of steps that keeps a team moving forward. However, this traditional approach offers almost no room for change if something unexpected occurs during the project.
- Testing is often delayed until the end of development. This could lead to the late discovery of issues or bugs, which can be costly and time-consuming to fix.
- The model does not incorporate mid-process feedback from users or clients, and changes are made based on results. This means that if a requirement changes or is misunderstood, the error might not be discovered until the project is near completion.
- All requirements must be known prior to development, which can greatly delay the project kick off. This can be challenging in projects where requirements may evolve or are not fully known at the start.
- The low flexibility level of the Waterfall model makes it difficult to make changes during development, and in some cases, it may even be impossible.

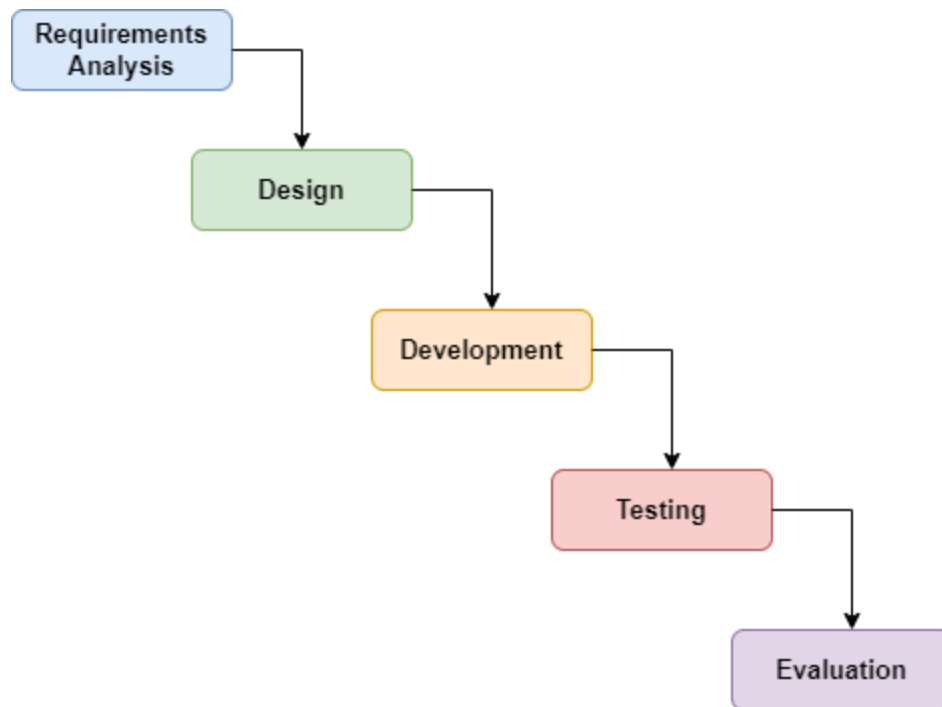


Figure 3: Waterfall Methodology (Ardhiansyah, et al., 2022)

3.1.2 Agile Methodology

Agile is a project management methodology that is iterative, introspective, and adaptive. In Agile practice, a project is divided into smaller sub-projects, commonly known as sprints. Upon the completion of each sprint, stakeholders and the team review the work done, make necessary adjustments for the upcoming sprint, and continue this process until the project is completed. The essence of Agile lies in the continuous, incremental delivery of value throughout the project's duration, rather than delivering it all at once at the end (Davis, 2022).

The agile methodology is not being selected for this project due to its inherent characteristics. The iterative nature of agile could potentially result in a project extending indefinitely, which is not suitable for this project that requires a defined timeline. Additionally, agile's sprint-based approach can make it difficult to estimate the total time, effort, cost, and resources at the start of a project. These factors combined make Agile a less desirable choice for this particular project.



Figure 4: Agile Methodology (Grepon, et al., 2021)

3.2 Selected Methodology

3.2.1 Evolutionary Prototyping

The Evolutionary Prototyping Model is a software development process where the prototype of a system is continually refined based on customer feedback until it evolves into the final product. This model is particularly useful when the customers do not know the exact project requirements beforehand.

The reason for selecting evolutionary prototyping for this project is that it perfectly aligned with the project's requirements and objectives. This approach allows for the continuous refinement of the system based on user feedback, ensuring that the final product is tailored to meet the users' needs effectively. It also facilitates the early detection and resolution of potential issues, thereby reducing the risk of project failure. Furthermore, evolutionary prototyping promotes active user involvement, fostering a sense of ownership and satisfaction among the stakeholders. Thus, this methodology not only enhances the quality of the deliverable but also ensures its acceptance and success in the target environment.

The advantages are:

- The tool can be refined based on user feedback, leading to a more effective and user-friendly tool.
- Since the prototype is tested at each iteration, errors can be detected and fixed early.
- The model is flexible, allowing new requirements to be easily accommodated on the project.
- Developers and users both understand the system better, which can lead to more effective network vulnerability assessments.

The disadvantages are:

- The model can be time-consuming due to the iterative nature of the process.
- The project may become complex due to continuous changes.
- There might be poor documentation because of continuously changing requirements.

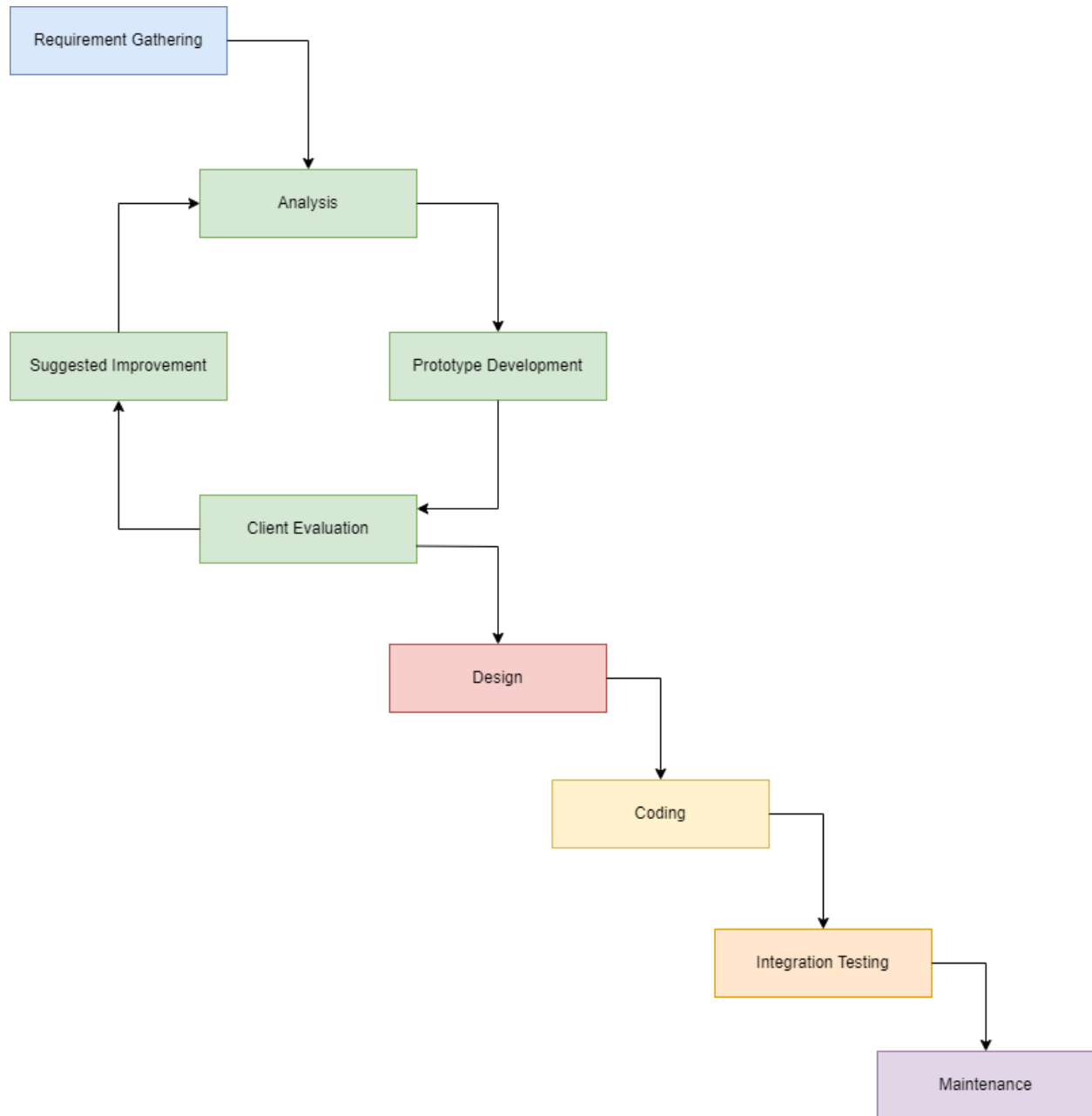


Figure 5: Evolutionary Prototyping (*Lapada, 2019*)

3.3 Work Breakdown Structure

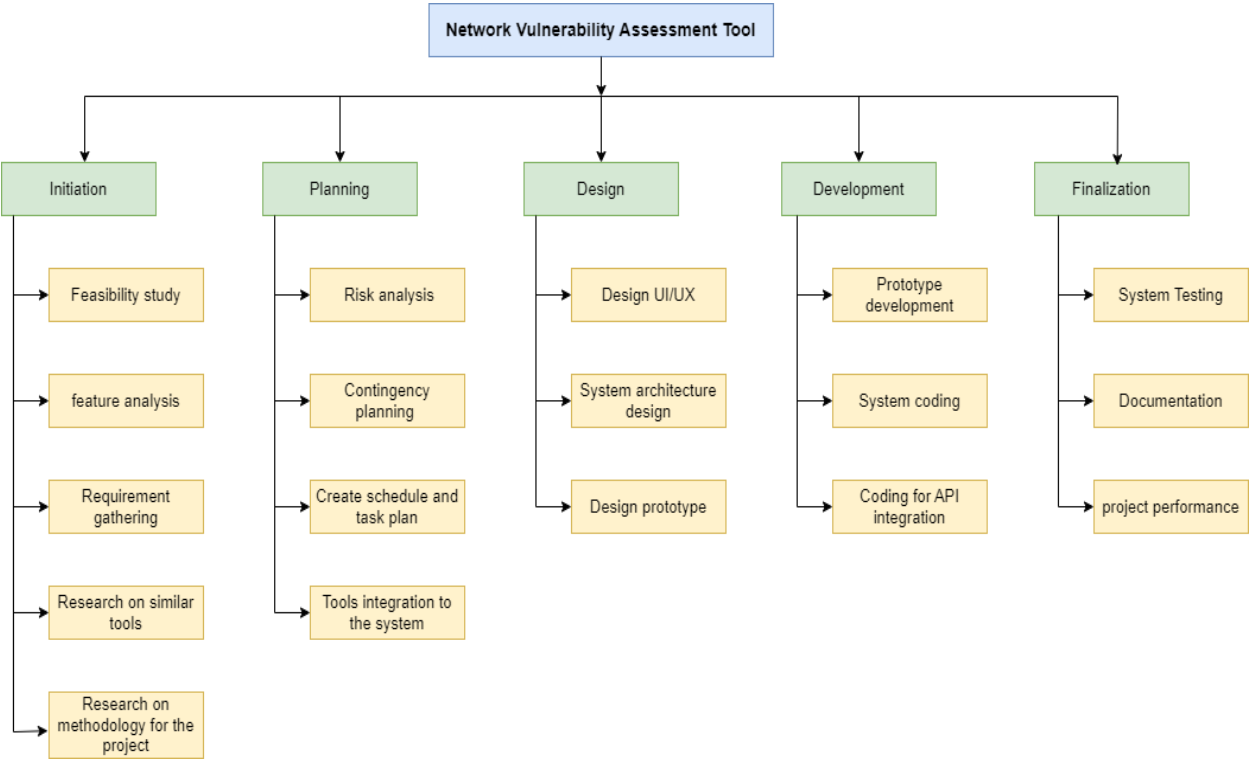


Figure 6: Work Breakdown Structure (WBS) of the project

3.4 Gantt Chart

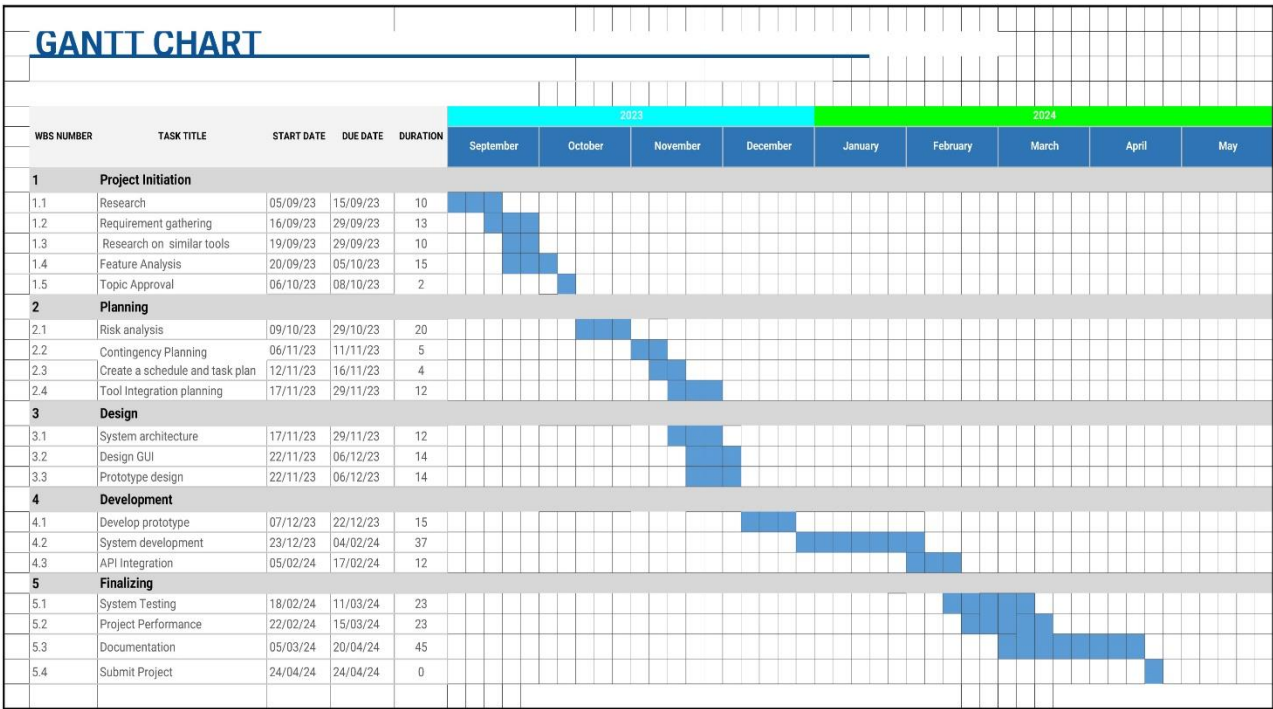


Figure 7: Gantt Chart of the project

4 Analysis

4.1 Progress Table

S.N.	Task Name	Start Date	Due Date	Status
1	Project Initiation			
1.1	Research	5/9/2023	15/09/2023	Completed
1.2	Requirement gathering	16/09/2023	29/09/2023	Completed
1.3	Research on similar tools	19/09/2023	29/09/2023	Completed
1.4	Feature Analysis	20/09/2023	5/10/2023	Completed
1.5	Topic Approval	6/10/2023	8/10/2023	Completed
2	Planning			
2.1	Risk Analysis	9/10/2023	29/10/2023	Completed
2.2	Contingency Planning	6/11/2023	11/11/2023	Completed
2.3	Creating a schedule and task plan	12/11/2023	16/11/2023	Completed
2.4	Tool integration planning	17/11/2023	29/11/2023	In Progress
3	Design			
3.1	System architecture	17/11/2023	29/11/2023	Completed
3.2	Design GUI	22/11/2023	6/12/2023	Completed
3.3	Prototype design	22/11/2023	6/12/2023	In progress
4	Development			
4.1	Develop prototype	7/12/2023	22/12/2023	In Progress
4.2	System development	23/12/2023	4/2/2024	To Do
4.3	API Integration	5/2/2024	17/02/2024	To Do
5	Finalizing			
5.1	System Testing	18/02/2024	11/3/2024	To Do
5.2	Project performace	22/02/2024	15/03/2024	To Do
5.3	Documentation	5/3/2024	20/04/2024	To Do
5.4	Submit project	24/04/2024	24/04/2024	To Do

Figure 8: Progress table of the project

4.2 Progress Review

The project is progressing systematically according to the detailed progress table. The project Initiation phase, which included tasks such as research, requirement gathering, research on similar tools, feature Analysis, and topic Approval, has been successfully completed. This phase laid the groundwork for the project by defining the scope and objectives.

The planning phase followed, with tasks like risk analysis, contingency planning, and creating a schedule and task plan being completed as planned. However, tool integration planning is still in progress. This phase is crucial as it involves outlining the project's roadmap and identifying potential risks. In the design phase, tasks such as system architecture and design GUI have been successfully completed, while prototype design is ongoing. This phase is where the theoretical plans start to take shape into a tangible model.

The development phase is currently underway, with the task of develop prototype in progress. However, system development and API Integration are yet to be started. This phase will involve the actual creation of the software tool. Lastly, the final phase, which includes tasks like system testing, project performance, and documentation, is also pending. This phase will ensure that the tool is ready for deployment and use. Overall, the project is advancing smoothly in accordance with the timeline outlined in the Gantt chart. The majority of the tasks are either already completed or currently underway.

Evidence of GUI design is in [appendix](#).

5 Future work

The tool is currently under development, with some features and functionalities already operational. However, there is ongoing work to enhance the system and ensure that all aspects of the tool are fully functional. This involves refining existing features, adding new ones, and ensuring that they all work seamlessly together to provide a comprehensive network vulnerability assessment. Another key area of future work involves integrating various APIs into the tool. This will enhance the tool's capabilities and ensure seamless interaction with other systems. API integration will allow the tool to pull in data from various sources, enhancing its ability to assess network vulnerabilities.

Once the tool's development and API integration are complete, comprehensive system testing will be conducted. This involves identifying and fixing any bugs or issues, ensuring the tool's reliability and efficiency. System testing is crucial to ensure that the tool works as expected and can effectively assess network vulnerabilities. Evaluating the project's performance is another important aspect of future work. This involves assessing how well the project has been executed, identifying areas of improvement, and optimizing operational efficiency. The insights gained from this evaluation will be used to refine the tool and improve future projects. Detailed documentation will be completed to provide users with guides and manuals for effectively utilizing the tool. This documentation will include step-by-step instructions on how to use the tool, explanations of the tool's features and functionalities, and troubleshooting tips. Advanced security protocols will be implemented to safeguard data integrity and privacy. As a network vulnerability assessment tool, it is crucial that the tool itself is secure and protects user data.

The next phase is to launch the project after getting feedback from the client and users, refining the project according to it, and focusing on small, medium, and large organization networks. This feedback will be instrumental in fine-tuning the tool to ensure it meets the unique needs of organizations of different sizes. The tool will be refined based on this feedback, with adjustments made to its features, functionalities, and user interface to ensure it is intuitive and effective for all users, regardless of the

size of their organization. Once these refinements are made, the project will move into the launch phase. This will involve a comprehensive rollout of the tool across organizations of varying sizes, accompanied by user training and support to ensure effective utilization.

The launch signifies the beginning of a new phase of continuous improvement and adaptation, ensuring that the tool remains a relevant and effective solution for network vulnerability assessment across all organization sizes. This ensures that the tool can provide a cost-effective solution for network vulnerability assessment for organizations of all sizes, thereby enhancing their overall cybersecurity posture.

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8 Appendix

8.1 Survey Findings

The survey of network vulnerability assessment tool project is conducted to gain insights from industry experts. The survey includes a variety of questions designed to understand the size and complexity of the respondents' networks, the types of devices and operating systems in use, and their familiarity with and usage of network vulnerability assessment tools. The survey also seeks to understand the features that respondents find most important in a network vulnerability assessment tool, such as comprehensive vulnerability scanning, detailed reporting, user-friendliness, and integration with other systems. It also asks about the frequency of network vulnerability assessments, the overall performance and speed of the tool, and the confidence in the accuracy of the vulnerability assessments provided by the tool.

Additionally, the survey inquires whether respondents have a dedicated IT security team or individual and what their budget is for a network vulnerability assessment tool. The responses to these questions will provide valuable insights into the current practices and needs in the industry, helping to guide the development of more effective and user-friendly network vulnerability assessment tools.

Due to privacy concerns, I am unable to provide the information of the individuals who responded to the survey form. Protecting the privacy and confidentiality of the respondents is my utmost priority. I adhere strictly to data protection regulations and ethical guidelines, which prohibit the sharing of any personal information of the respondents. The responses collected from the survey are used solely for the purpose of research and analysis.

How large is your network?

10 responses

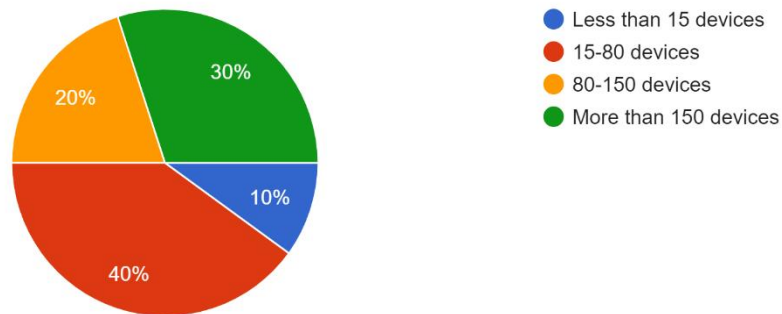


Figure 9: Survey Question 1

What types of devices are on your network? (Select all that apply)

10 responses

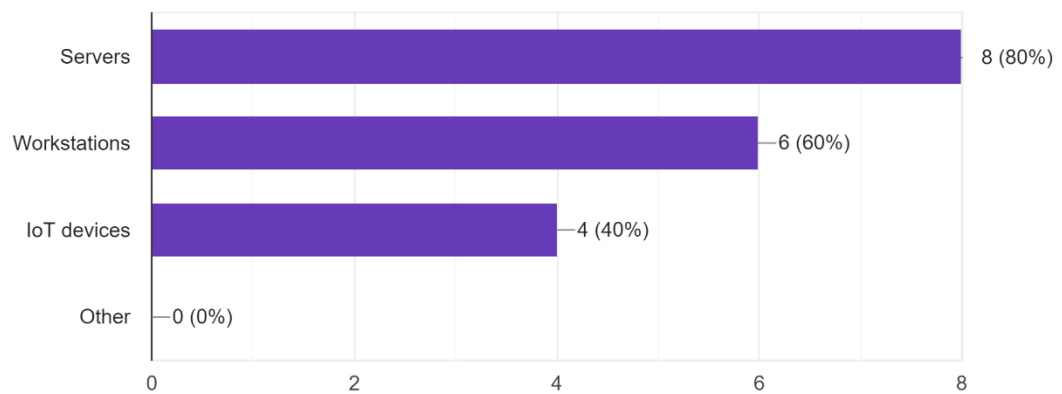


Figure 10: Survey Question 2

Which operating systems are in use on your network? (Select all that apply)

10 responses

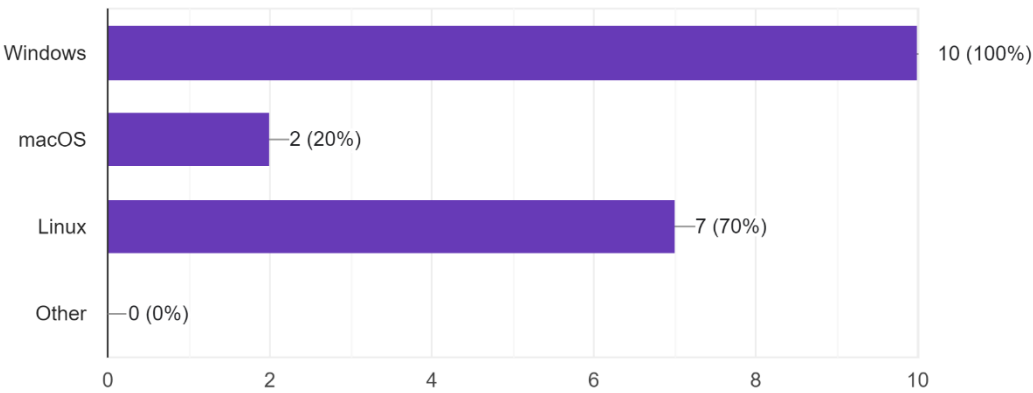


Figure 11: Survey Question 3

How familiar are you with network vulnerability assessment tools?

10 responses

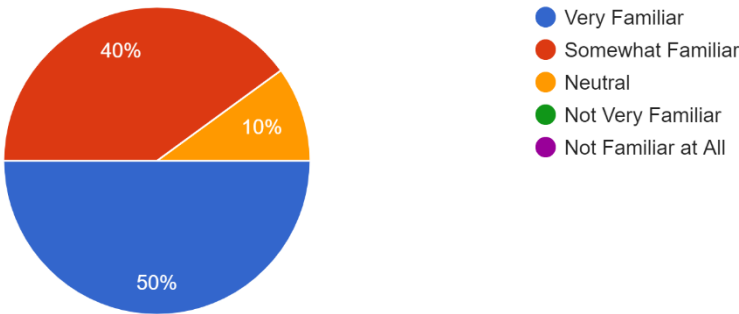


Figure 12: Survey Question 4

Do you currently use any network vulnerability assessment tools?
10 responses

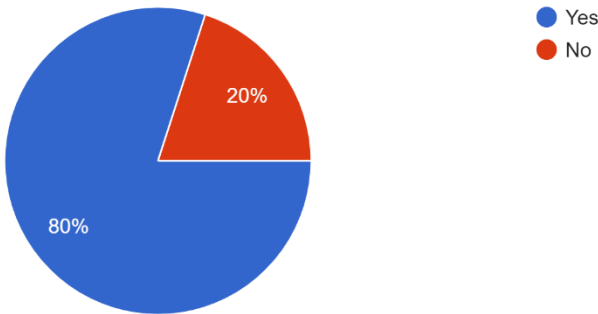


Figure 13: Survey Question 5

If yes, which network vulnerability assessment tool(s) do you use? (Select all that apply)
10 responses

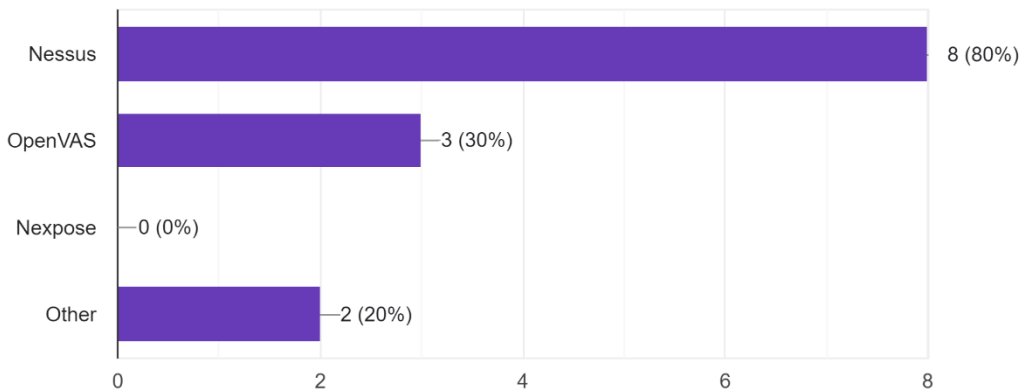


Figure 14: Survey Question 6

Which features do you find most important in a network vulnerability assessment tool? (Select all that apply)

10 responses

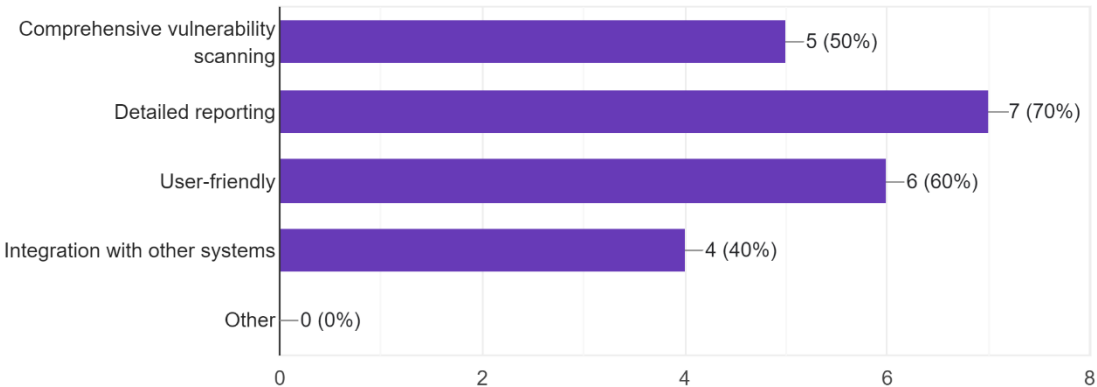


Figure 15: Survey Question 7

How frequently do you perform network vulnerability assessments?

10 responses

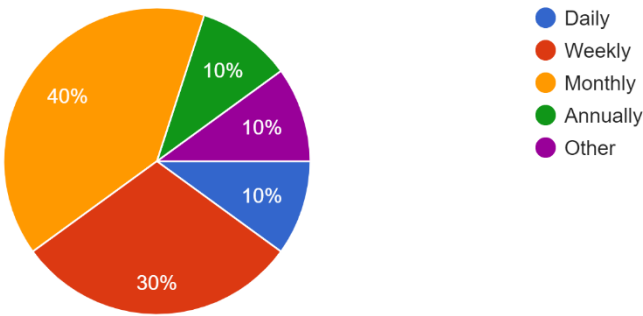


Figure 16: Survey Question 8

How would you rate the overall performance and speed of the tool?

10 responses

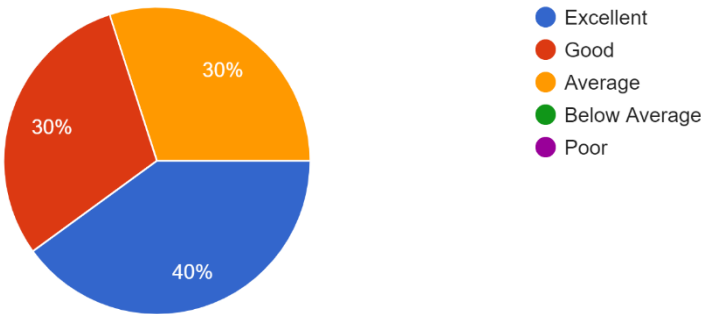


Figure 17: Survey Question 9

How confident are you in the accuracy of the vulnerability assessments provided by the tool?

10 responses

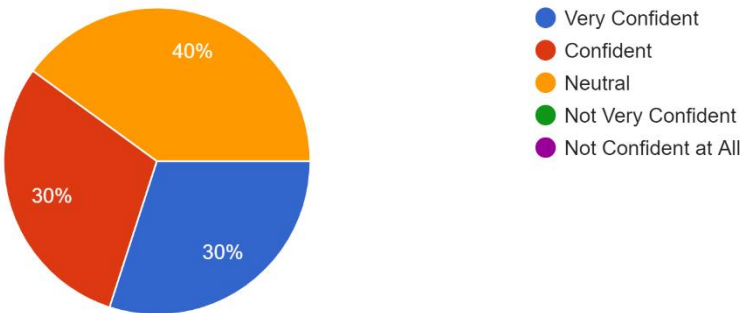


Figure 18: Survey Question 10

Do you have a dedicated IT security team or individual?
10 responses

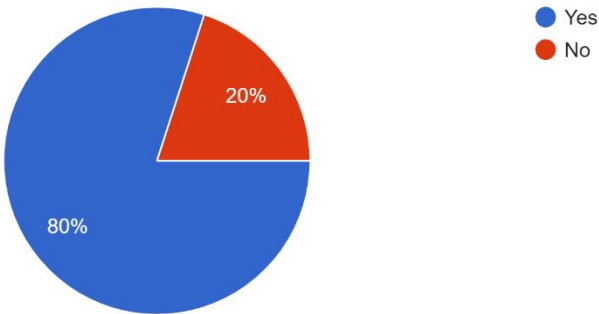


Figure 19: Survey Question 11

What is your budget for a network vulnerability assessment tool?
10 responses

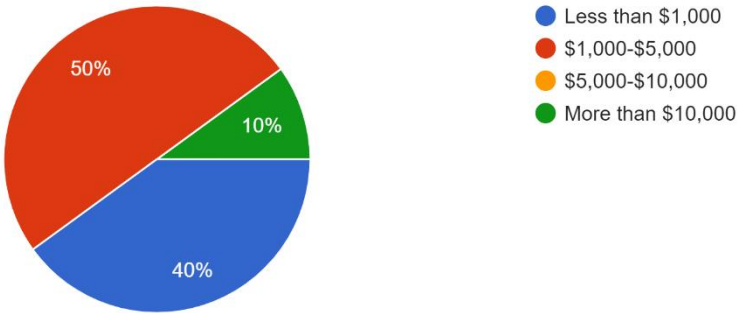


Figure 20: Survey Question 12

8.2 Milestone

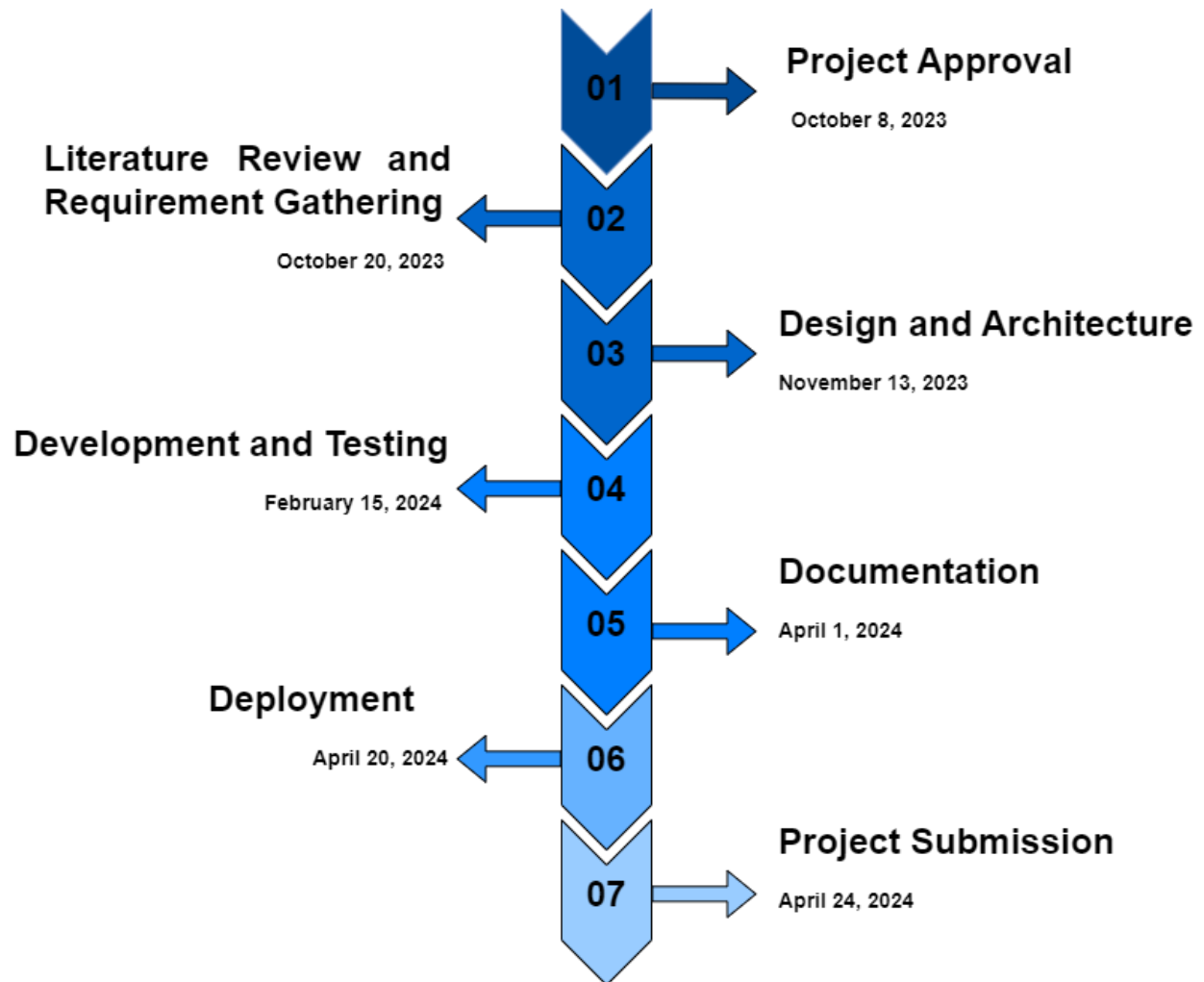
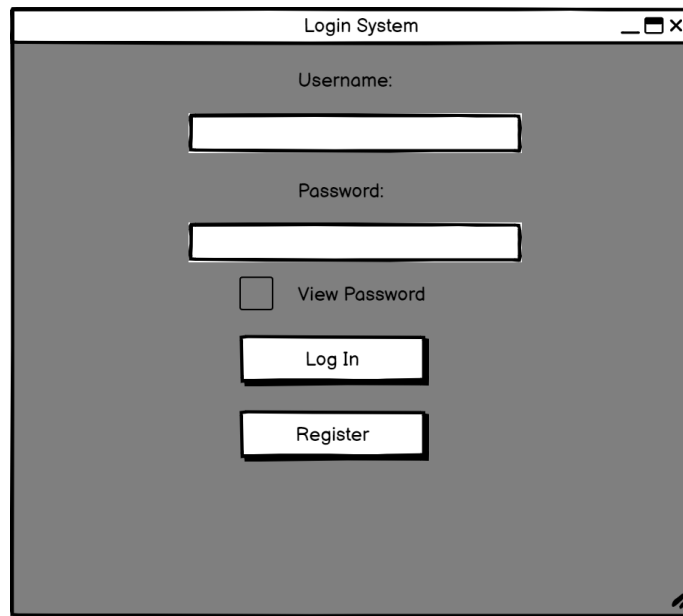


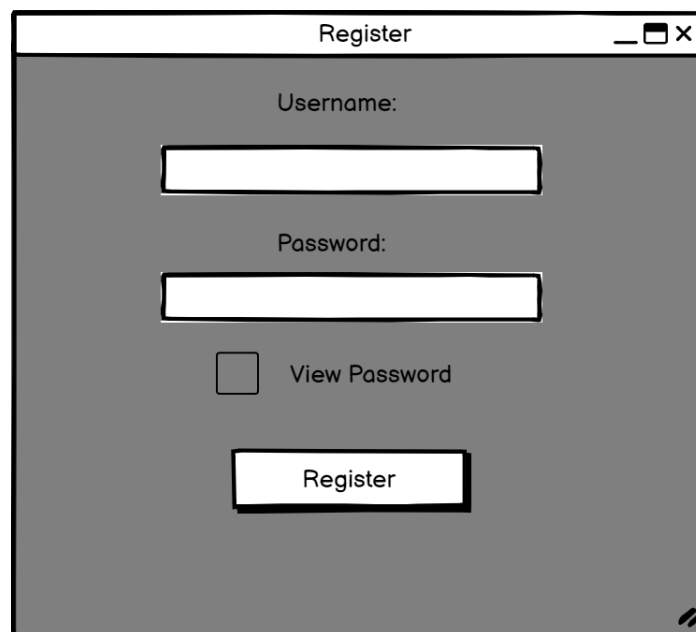
Figure 21: Milestone

8.3 Project's GUI Design



A screenshot of a GUI window titled "Login System". The window has a gray background and a white title bar with standard window controls. It contains two text input fields: the first is labeled "Username:" and the second is labeled "Password:". Below the password field is a checkbox labeled "View Password". At the bottom, there are two buttons: "Log In" and "Register".

Figure 22: Login Design



A screenshot of a GUI window titled "Register". The window has a gray background and a white title bar with standard window controls. It contains two text input fields: the first is labeled "Username:" and the second is labeled "Password:". Below the password field is a checkbox labeled "View Password". At the bottom, there is a single button labeled "Register".

Figure 23: Register Design

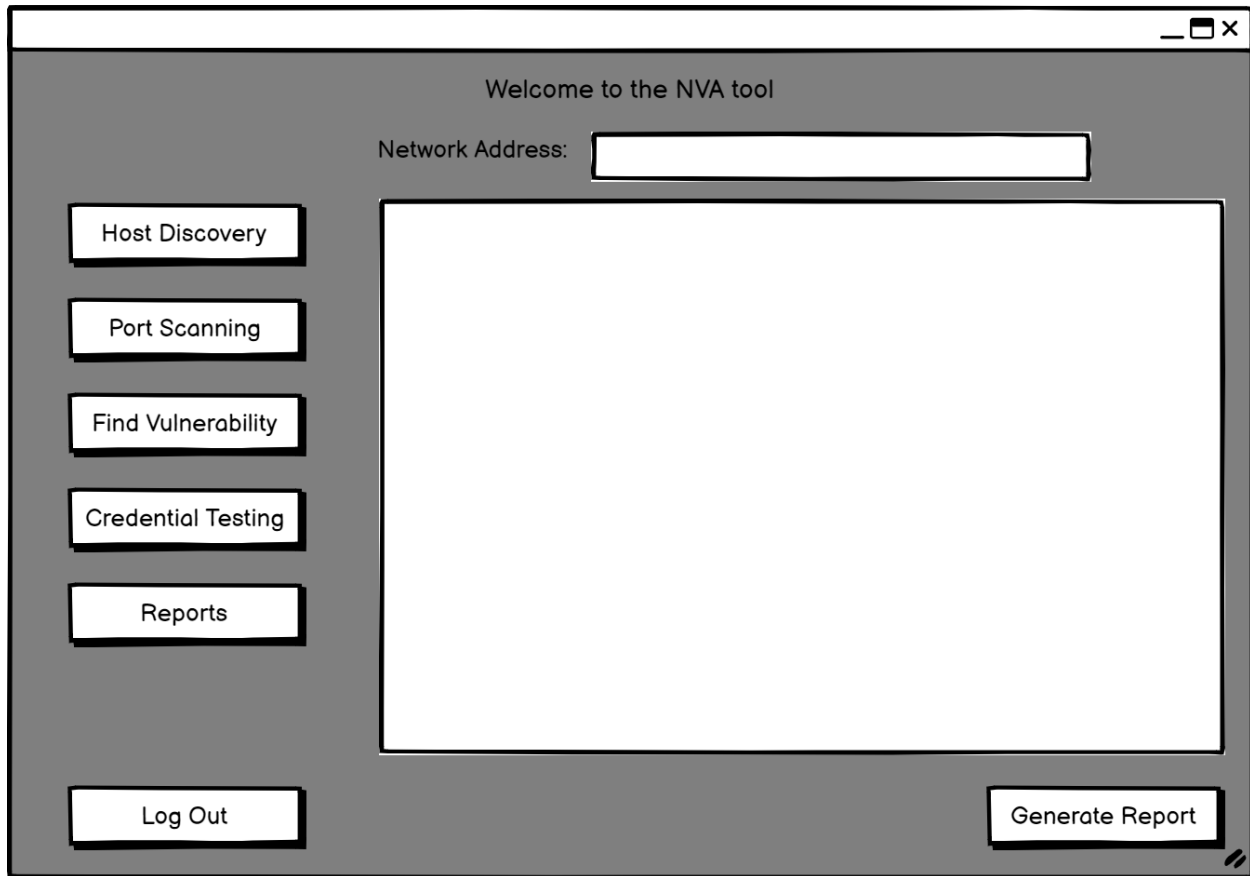


Figure 24: Dashboard Design

Back to previous [page](#).

8.4 Development

8.4.1 Login GUI

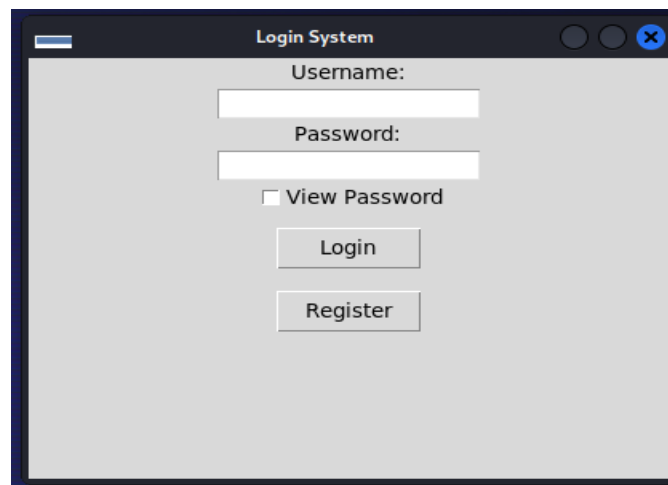
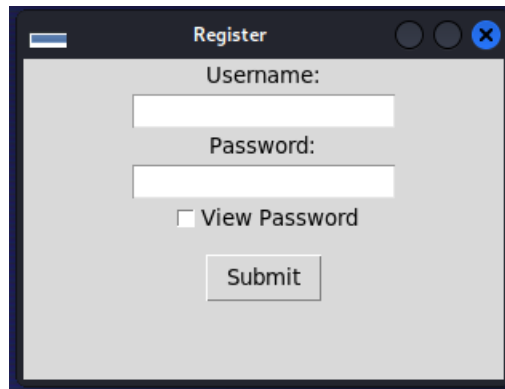


Figure 25: Login Interface

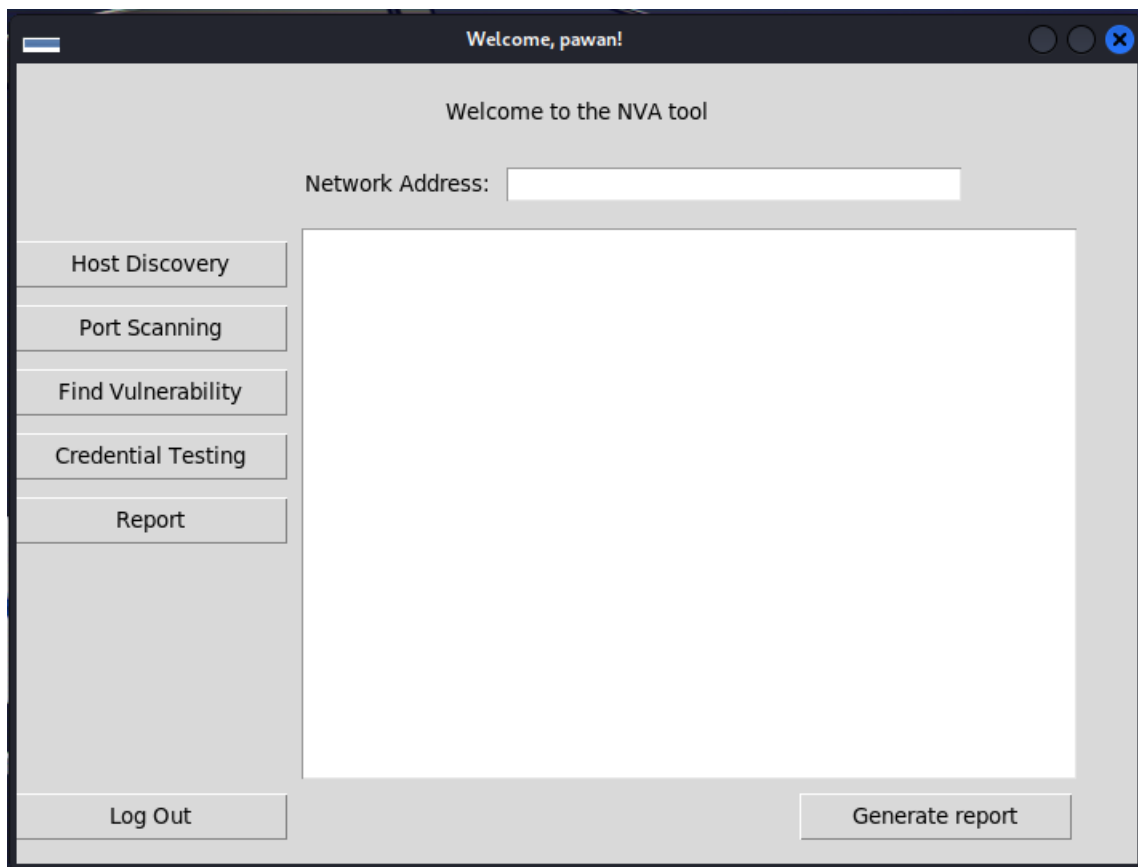
8.4.2 Register GUI



A screenshot of a 'Register' window. The window has a title bar with the text 'Register' and standard window controls (minimize, maximize, close). The main area is light gray and contains the following elements: a label 'Username:' followed by a text input field; a label 'Password:' followed by a text input field; a checkbox labeled 'View Password'; and a 'Submit' button at the bottom.

Figure 26: Register Interface

8.4.3 Dashboard GUI



A screenshot of a 'Dashboard' window. The window has a title bar with the text 'Welcome, pawan!' and standard window controls. The main area is light gray and contains the following elements: a label 'Welcome to the NVA tool'; a label 'Network Address:' followed by a text input field; a vertical sidebar on the left with buttons for 'Host Discovery', 'Port Scanning', 'Find Vulnerability', 'Credential Testing', 'Report', and 'Log Out'; and a 'Generate report' button at the bottom right. A large white rectangular area is positioned to the right of the sidebar buttons.

Figure 27: Dashboard Interface

8.5 Development Testing

8.5.1 Registering Username and Password

Creating new account using username as “admin123” and password as “admin123”.

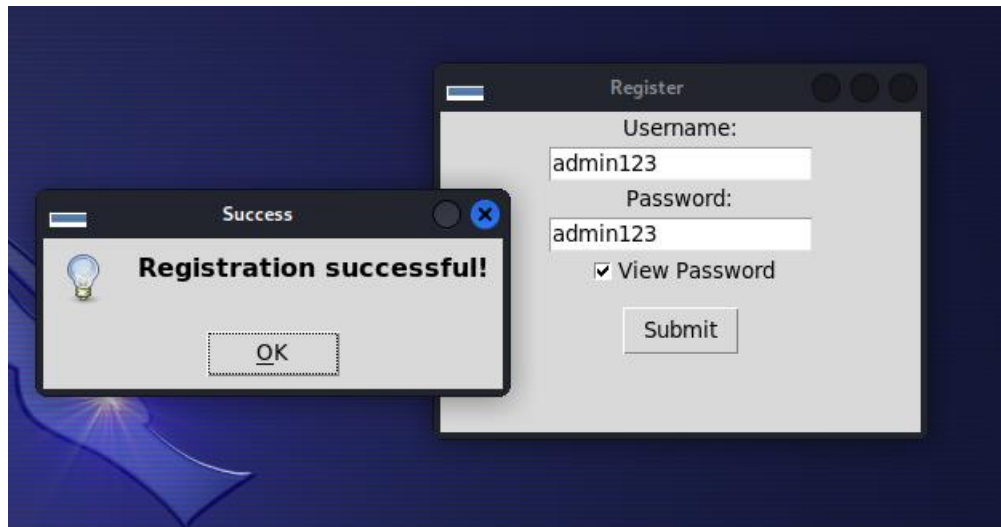


Figure 28: Registering Username and Password

8.5.2 Login

Login using recently created user account.

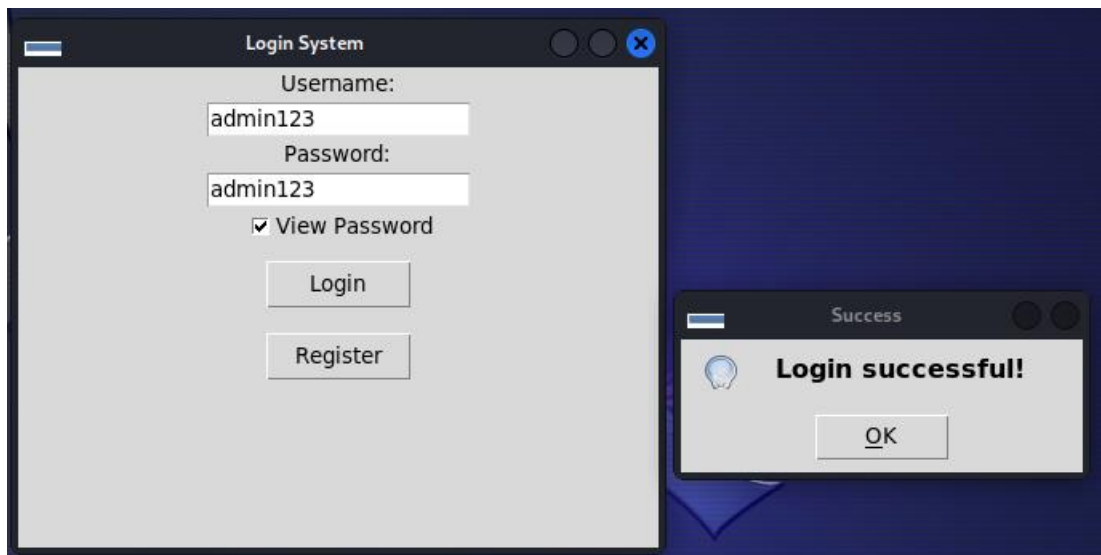


Figure 29: Login Using Login Credentials

8.5.3 Host Discovery Button

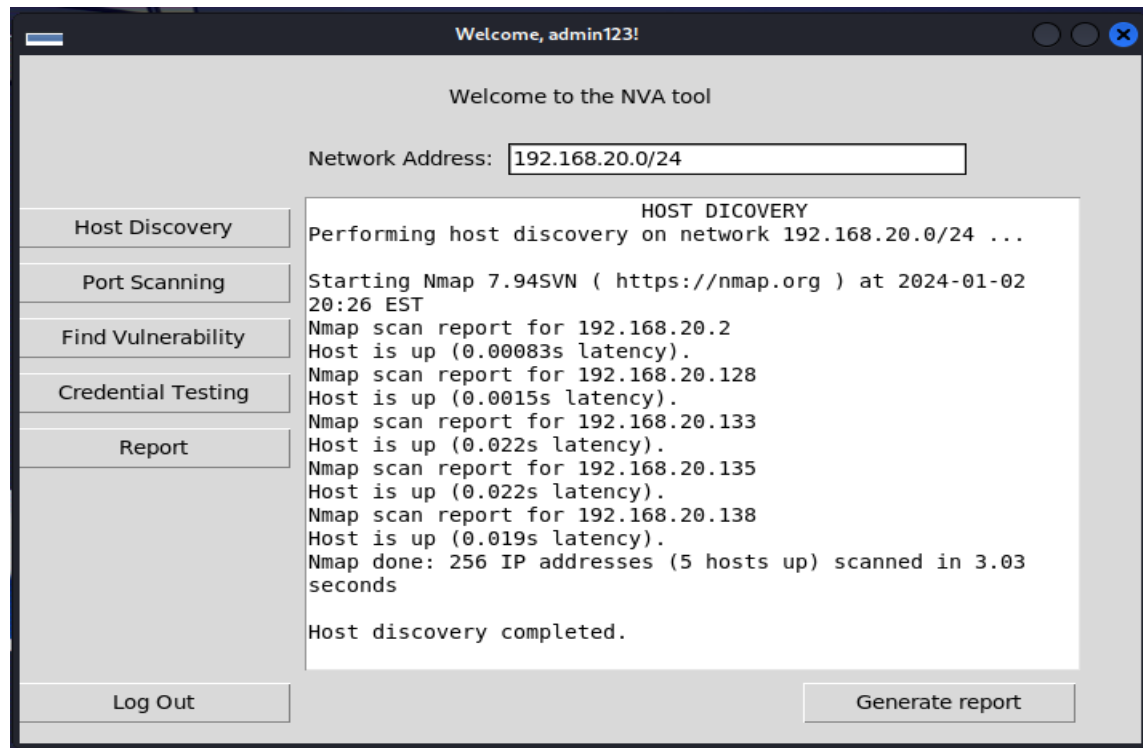


Figure 30: Host Discovery on the Network

8.5.4 Port Scanning Button

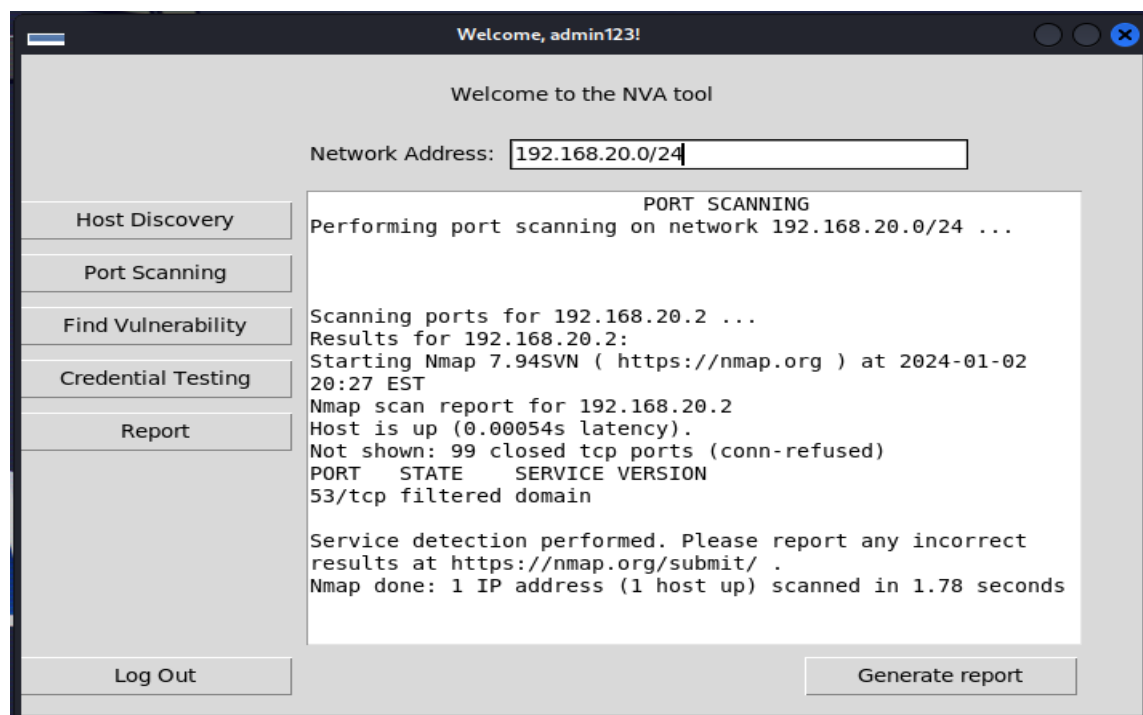


Figure 31: Port Scanning on the Network

8.5.5 Generate Report Button

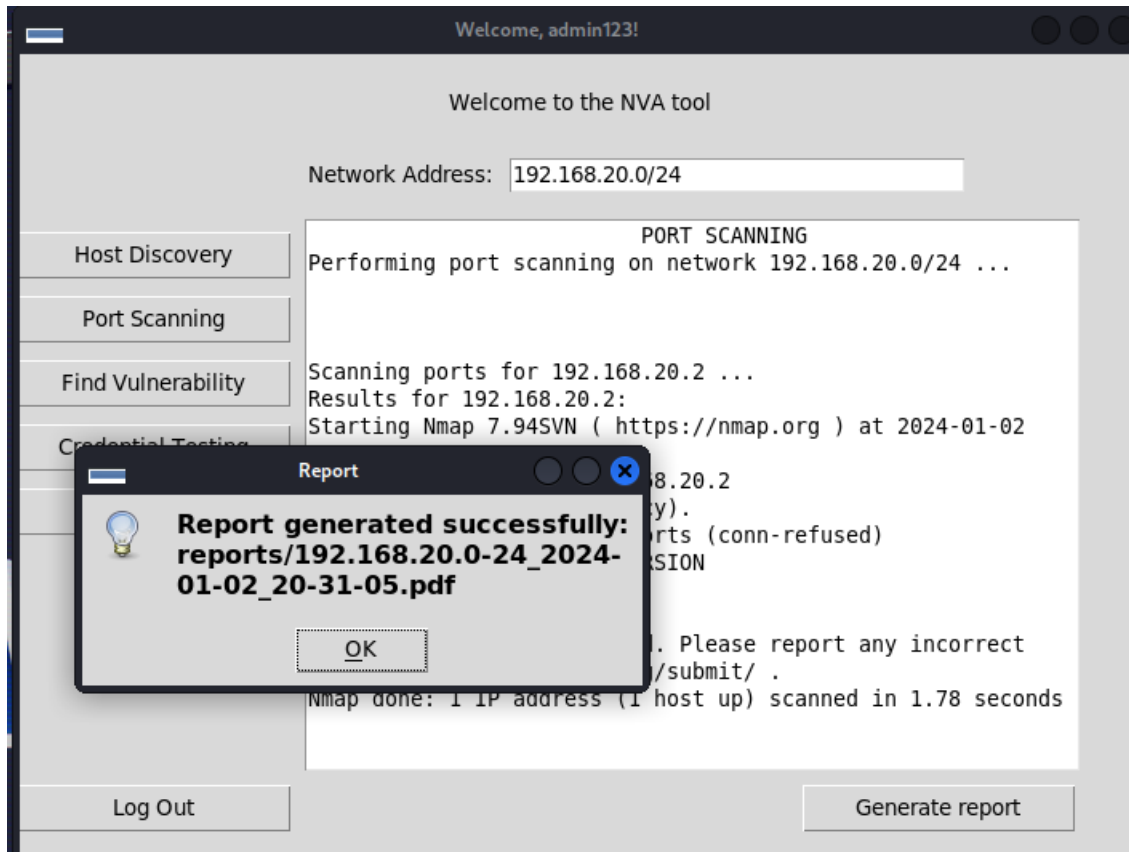


Figure 32: Generating Report

8.5.6 Report Button

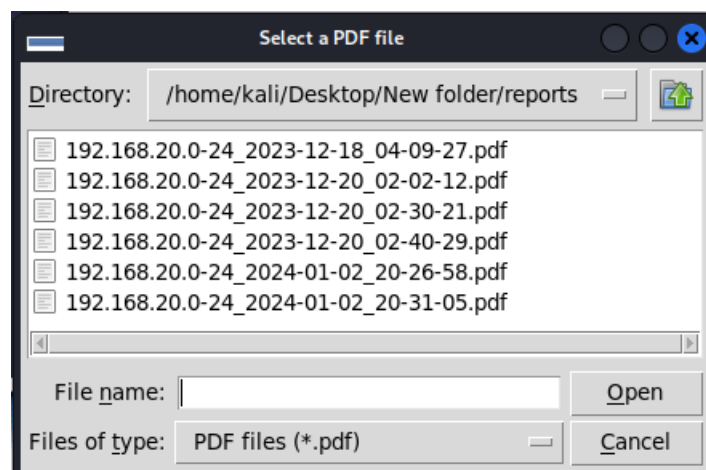


Figure 33: Report Lists

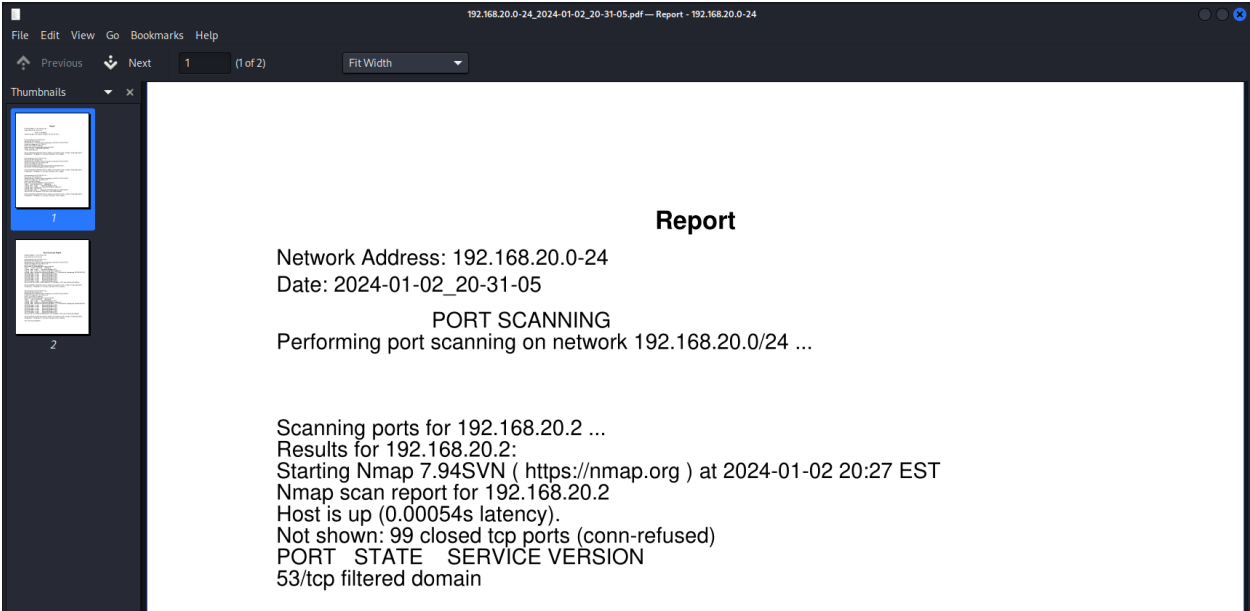


Figure 34: Pdf File

8.6 Client approval Letter



Figure 35: Client Approval Letter